RAFAEL HARRY GOMAR DAWSON

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Summary

Computer Engineering student at ITAM, focused on representation learning, NLP, and applied machine learning. Experienced in leading interdisciplinary research and building practical systems for law, public policy, and mobility.

Experience

Mantis – MIT CSAIL

Collaborating Researcher 08/2025 - Present

- Led interdisciplinary research on representation learning and embedding-based methods across law, patents, political science, and other applied domains.
- Served as an instructor and lecturer for MIT IAP 6.S189, 6.S187, and MIT Sloan 15.S65, covering applied AI, machine learning, and deployment.

MAKERS – ITAM

Coordinator, Computer Science Division 08/2023 - Present

- Led and mentored 15 students in 7 development and 9 research projects that integrate computing to benefit the ITAM community, fostering collaboration and technical growth.
- Oversaw recruitment, organization, and participation across the division's 2 areas (Data science and Software Development).

Universo de Leyes Initiative

Lead Researcher 11/2022 - 2025

- Developed an NLP-based information extraction system that generates knowledge graphs to visually represent citational relationships of Mexican legal domains.
- Created a comprehensive annotated dataset to fine-tune a RoBERTa model and train a named entity recognition pipeline using SpaCy. Authored a research paper detailing findings; presented at ICITS 2025.

Spingere

Research Intern 06/2022 - 08/2022, 06/2023 - 07/2023

- Conducted research, planning and prototyping of an NLP solution to automate COSMIC software functional size analysis.

Awards & Achievements

ITAM Hackathon	**ISAC Hackathon**	**INEGI Datatón**
Winner, 2023	Winner, 2025 | Finalist, 2024	Winner, 2025 | Finalist, 2024

ITAM Engineering Academic Excellence Recognition | 2023, 2024, 2025

Relevant Projects

Tellus — Geospatial ATM Expansion Planning

Built a geospatial platform that ranks ATM sites for three major banks using local demand, competitor coverage, transaction potential, and learned location representations.

Retail Informality Prioritization Model — INEGI Datatón 2025

Won INEGI Datatón 2025 with an analysis that pinpointed the products, regions, and business segments where policy intervention could have the greatest effect on commercial informality.

- Built a prioritization model from wage, social-security, education, and Economic Census data to diagnose barriers to formalization and match them with targeted policy actions.

SEMOVI — Public Transport Route Digitization

Built a pipeline for Mexico City's mobility authority, SEMOVI, that converts written public-transport route records into digital map data.

- Processed official scanned documents, extracted route descriptions, and reconstructed the corresponding geospatial paths with minimal manual input.

Sustainable Auto Parts Matching

Worked with InvierteSustentable and JetzApp on a catalog-matching system that helps buyers find compatible, lower-impact alternatives to automotive parts.

- Prototyped LLM-based agents that adapt Python preprocessing functions to inconsistent catalog schemas and match equivalent parts across suppliers.

Publications & Seminars

Comparative Analysis of Information Extraction Techniques in the Mexican Legal System, F. Valdés, R. H. Gomar and D. A. Méndez

- Published in Proceedings by Springer in Lecture Notes in Networks and Systems 2025 and presented at the 8th International Conference on Information Technology & Systems, held at Politécnico Nacional

Universo de Leyes: Aplicando el Procesamiento del Lenguaje Natural para vincular textos jurídicos con un enfoque integral

- Seminario de Ingeniería de Software y Sus Implicaciones, Nov 2024, Facultad de Ciencias, UNAM
- Seminario Recursivo, Mar 2024, División Académica de Ciencias de la Computación, ITAM

Y si pudiéramos entender el contexto a través de los datos? | Player2Vec

- Hudl Statsbomb, ITAM Sports Analytics Conference 2025